

UNIT-5: Complex Variable – Differentiation

CO1: The students will learn the essential tool of Complex Variable – Integration in a comprehensive manner.

PART – A (5*2=10)

- A1:** Define a line integral of a complex function.
- A2:** What is meant by a contour in the complex plane ?
- A3:** State Cauchy–Goursat theorem.
- A4:** Define an isolated singularity of a complex function.
- A5:** What is a zero of an analytic function ?

PART – B (3*7=21)

- B1:** Evaluate the line integral $\int_C (z^2 + 1)dz$ where C is the straight line joining $z = 0$ to $z = 1 + i$.
- B2:** Calculate $\int_C \frac{z^2}{z-1} dz$, where C: $|z-1|=2$.
- B3:** Find the zeros of the function $f(z) = z^2 - 4$ and state their order.

PART – C (3*11=33)

- C1:** Evaluate the contour integral $\int_C \frac{e^z}{z-2} dz$, where C is the circle $|z-2| = 1$, using Cauchy Integral

Formula.

- C2:** Let $f(z) = 3 + 2i$. Determine whether $f(z)$ satisfies the conditions of Liouville's theorem and state the nature of the function.
- C3:** Define the Maximum Modulus Theorem and determine whether the function $f(z) = z^2$ can attain its maximum value inside the region $|z| < 1$.

Answers

B1: $\frac{(1+i)^3}{3} + (1+i)$.

B2: $2\pi i$.

B3: Zeros are $z = \pm 2$, each of order one.

C1: $2\pi i e^2$.

C2: $f(z)$ satisfies the conditions of Liouville's theorem because $f(z)$ is entire and bounded function and the nature of the function is constant.

C3: The maximum value is attained on the boundary $|z| = 1$, not inside the region.